

Akila Premathilaka

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🎓 Google Scholar

PROFILE

My research interests lie in Machine Learning and Deep Learning with a focus on Explainable AI (XAI). I am also deeply passionate about Time Series forecasting, Image Processing, and the practical applications of Large Language Models (LLMs). I am passionate about leveraging these techniques to build trustworthy, privacy-preserving models for real-world applications.

EDUCATION

University of Peradeniya <i>Bachelor of Science (Honors) in Data Science</i> • Last two years GPA: 3.62 / 4 • Overall GPA: 3.48 / 4.00 • Data Science GPA: 3.73 / 4.00	Peradeniya, Sri Lanka 2020 – 2025
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RESEARCH

Smart Diagnosis of Anemia Subtypes Using CBC and Deep Learning [PDF] [DOI] <i>Springer Nature Communications in Computer and Information Science (CCIS)</i> , Vol. 2726, pp. 135–149 Akila Premathilaka*, Chamila Walgampaya	Published 2025
Assessment of Floating Photovoltaic Solar Energy Potential on Major Reservoirs in Sri Lanka: A Physical Modelling Approach	Under Review
S.D.A.V.S. Premathilaka*, R.D. Yapa, R. Punchi-Manage	
Uncovering Solar Energy Potential of Reservoirs in Sri Lanka [PDF] [DOI] <i>iPURSE 2025 Proceedings (Conference Abstract)</i> S.D.A.V.S. Premathilaka*, R.D. Yapa, R. Punchi-Manage	Published 2025
Machine Learning for Anaemia Subtype Classification [PDF] [DOI] <i>iPURSE 2025 Proceedings (Conference Abstract)</i> S.D.A.V.S. Premathilaka*, C.K. Walgampaya	Published 2025
Assessing Floating Solar Potential using SARIMA-LSTM [PDF] [Link] <i>RESCON 2025 Proceedings (Conference Abstract)</i> S.D.A.V.S. Premathilaka*, R.D. Yapa and R. Punchi-Manage	Published 2025
Forecasting Solar Energy using LSTM Models [PDF] [DOI] <i>IRC-OUSL 2025 Proceedings (Conference Abstract)</i> S.D.A.V.S. Premathilaka*, R.D. Yapa and R. Punchi-Manage	Published 2025

PROJECTS

Large Language Model Approach for Tabular Data Classification • Developing a new framework that uses Large Language Models (LLMs) to classify tabular data by converting tables into various textual serialization formats. This project explores prompt-based and embedding-driven methods to capture semantic context. The goal is to achieve better flexibility and generalization than traditional machine learning models.	Ongoing
Visualization and Analysis of major river water levels in Sri Lanka [Github] • Developed an interactive dashboard to visualize river water levels and monitor flood risks across major rivers in Sri Lanka. The system integrated predictive modelling for time-series forecasting, using upstream water level data to provide early flood warnings. This improved situational awareness and supported data-driven decision-making for disaster management.	

Dimensionality Reduction Using Factor Analysis on a Diabetes Dataset [Github]

- Conducted factor analysis on a diabetes dataset using R to identify latent variable structures and reduce feature dimensionality. Applied Exploratory (EFA) and Confirmatory (CFA) Factor Analysis, validating factor suitability via Scree Plots and Bartlett's Test. This optimized feature selection while successfully minimizing overfitting and multicollinearity to improve overall model interpretability.

Canonical Correlation Analysis of Car Performance Data [Github]

- Performed Canonical Correlation Analysis (CCA) using R to examine complex relationships between multivariate vehicle specifications and performance characteristics. Conducted thorough preprocessing and exploratory data analysis, evaluating the statistical significance of canonical dimensions to identify key contributing variables. The analysis successfully derived critical insights to optimize vehicle efficiency and design.

EXPERIENCE

Temporary Instructor (Teaching Assistant)

May 2025 – Present

Department of Computer Engineering, University of Peradeniya

Kandy, Sri Lanka

- Organized, created and conducted lab and tutorial sessions for undergraduate modules:
Programming for Engineers I & II (CO1010 – Instructor In Charge), Data Structures and Algorithms I & II (CO1030 – Instructor In Charge), Computer Systems Programming (CO1020), Programming Languages (CO523 – Instructor In Charge), Data Mining & Machine Learning (CO544), Neural Network & Deep Learning (CO5420), Image Processing (CO5430), Database Systems (CO2050), Network and Web Application Design (CO324)

Intern Data Scientist

Oct 2024 – Apr 2025

OCTAVE, John Keells Holdings PLC

Colombo, Sri Lanka

- Developed a backend system for a travel agent platform with real-time hotel recommendations, vehicle selection, and budget-based cost calculations.
- Built a customer classification model for a hotel chain to segment users into behavioral clusters, enabling personalized marketing and increased bookings.
- Analyzed Retail Media Network (RMN) campaign requirements and generated targeted customer segments using demographic and behavioral data.
- Automated a data pipeline to evaluate campaign performance and generate actionable insights for marketing optimization.

OUTREACH & LEADERSHIP

Membership Management Officer

Mar 2024 - Nov 2024

DataEx - Data Science Society, University of Peradeniya

Kandy, Sri Lanka

Executive Committee Member

Mar 2024 - Nov 2024

DataEx - Data Science Society, University of Peradeniya

Kandy, Sri Lanka

Member

Jan 2021 - Jan 2025

Computer Society, University of Peradeniya

Kandy, Sri Lanka

Member

Jan 2021 - Jan 2025

Statistical Circle, University of Peradeniya

Kandy, Sri Lanka

TECHNICAL SKILLS

Programming Languages:

Java, Python, C, R, HTML/CSS

Database Management Systems:

MySQL, PostgreSQL, MongoDB, Neo4j

Big Data Platforms:

Databricks, DBFS

Version Control:

Git

CERTIFICATIONS

- Natural Language Processing with Classification and Vector Spaces [Coursera]
- Fundamentals of AI Agents Using RAG and LangChain [Coursera]
- Supervised Machine Learning: Regression and Classification [Coursera]
- Introduction to Data Engineering [Coursera]
- Introduction to Responsible AI [Google Cloud Skills Boost]
- Introduction to Large Language Models [Google Cloud Skills Boost]
- Databases and SQL for Data Science with Python [Coursera]

SELECTED COURSEWORK

- **Data Science & Computer Science:**
Artificial Intelligence and Expert Systems, Artificial Neural Networks and Fuzzy Logic, Machine Learning, Big Data Analytics, Computing for Data Science, Advanced Database Management Systems, Data Structures, Design and Analysis of Algorithms, Computer Architecture
- **Statistics:**
Probability Theory, Regression Analysis, Multivariate Methods, Advanced Time Series Analysis, Data Mining Techniques, Nonparametric and Categorical Data Analysis, Reliability Theory and Survival Analysis, Statistical Simulations, Quality Control
- **Mathematics:**
Linear Algebra, Calculus, Numerical Analysis, Set Theory

REFERENCES

Prof. C. K. Walgampaya (PhD), Professor, Department of Engineering Mathematics, Faculty of Engineering, University of Peradeniya, Sri Lanka
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Dr. Erunika Dayaratne (PhD), Senior Lecturer, Department of Statistics & Computer Science, Faculty of Science, University of Peradeniya, Sri Lanka
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